

RESEARCH PAPER

Translation and Cultural Adaptation of the GEQ into Brazilian Portuguese

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Abstract. This study describes the process and results of the translation and cultural adaptation of the Game Experience Questionnaire (GEQ) into Brazilian Portuguese, an instrument designed to evaluate game experience. The methodological process included translation, back-translation, expert committee review, and pilot testing, assessing item comprehension, internal consistency, cultural adequacy, and semantic and conceptual equivalence. The study was conducted with 115 university students (18–24 years old) playing playing education- and health-related serious games developed at the host university, and confirmed internal consistency and conceptual equivalence with the original instrument through Exploratory Factor Analysis (EFA), with Cronbach’s Alpha coefficients, $\alpha > 0.8$ in most cases, and McDonald’s Omega $\omega > 0.9$, demonstrating adequate factorial structure and high reliability. Results indicate that the adapted GEQ version proved valid and reliable for assessing game experience in the Brazilian context. This study contributes to advancing game research in Brazil, enabling cross-cultural comparisons and supporting the development of games better aligned with local preferences.

Keywords: Translation; User Experience; Cultural Adaptation; Game Experience Questionnaire

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1 Introduction

In recent years, digital games have gained increasing relevance across diverse fields of knowledge, transcending their initial entertainment purpose to become valuable tools in educational, therapeutic, and training contexts. User experience (UX) during interaction with digital games is a critical factor for the success of such applications, directly influencing the effectiveness and acceptance of games by their target audience [da Silva *et al.*, 2024]. In this scenario, the Game Experience Questionnaire (GEQ), originally proposed by Poels, de Kort, and IJsselsteijn [2007], has emerged as one of the most robust and widely used instruments for evaluating game experience, enabling the measurement of dimensions such as immersion, flow, competence, positive and negative affect, tension, and challenge. The GEQ has been applied in international studies [Zulfa *et al.*, 2020; IJsselsteijn *et al.*, 2008; Sabet *et al.*, 2019] to investigate player experience, ranging from entertainment games to serious health interventions.

Despite its widespread use, the absence of a validated version of the GEQ in Brazilian Portuguese represents a significant gap in the literature. Cross-cultural adaptation of research instruments is a complex process that extends well beyond mere linguistic translation, requiring careful consideration of cultural, social, and contextual nuances that may affect respondents’ interpretation of items [Brislin, 1970]. In the Brazilian context, marked by significant cultural and linguistic diversity, adapting the GEQ presents a considerable yet indispensable challenge. Although Brazil has a substantial volume of research on usability and UX in digital games [Oliveira da Silva Junior *et al.*, 2024], there is a lack

of methodologically instruments adapted to the sociocultural and linguistic particularities of the Brazilian population. This methodological gap can compromise the ecological validity and accuracy of results, reinforcing the need for studies that address the cross-cultural adequacy of game evaluation tools [Oliveira da Silva Junior *et al.*, 2024].

The translation and cross-cultural adaptation of psychological scales, including those used to evaluate game experience, are critical steps to ensure the validity and comparability of results across different cultural and linguistic contexts. A simple literal translation may be insufficient, as concepts and expressions may carry different meanings or cultural nuances that affect participants’ comprehension and responses. The absence of an adequate cross-cultural adaptation can lead to [Oliveira da Silva Junior *et al.*, 2024]:

- **Cultural Bias:** Culture-specific items may not be understood in the same way across cultures, leading to biased responses.
- **Lack of Conceptual Equivalence:** The construct that the scale intends to measure may not be interpreted identically across cultures, compromising construct validity.
- **Unreliable Results:** Lack of linguistic and cultural equivalence can yield inconsistent and unreliable data, making valid cross-group comparisons impossible.

The cross-cultural adaptation process typically involves multiple stages, including back-translation, expert review, and pre-testing with the target population. This methodological rigor is essential to ensure that the translated version retains the psychometric properties of the original, measuring the same construct reliably and validly in the new cultural

context. The importance of this process is further underscored by the observation that game experience measures frequently lack empirical validation, and most focus on motivation or enjoyment rather than acquired skills [Williams *et al.*, 2023]. Furthermore, due to the immersive and culturally influenced nature of games, how players perceive and respond to narrative, graphics, mechanics, and social interactions is deeply shaped by their cultural experiences. For game experience questionnaires such as the GEQ to be effective in international studies, their translated versions must be rigorously adapted to reflect the cultural and linguistic nuances of each population.

The adaptation and validation of the GEQ for the Brazilian context will contribute to the advancement of digital game research in the country, enabling more robust comparisons in international studies and ensuring greater reliability and applicability of data. This article therefore aims primarily to describe the process of translating and culturally adapting the GEQ into Brazilian Portuguese, highlighting the strategies adopted to ensure the validity and reliability of the adapted instrument.

2 Literature Review

The cultural adaptation of research instruments is a complex methodological process requiring linguistic translation alongside attention to sociocultural, cognitive, and contextual aspects that may influence respondents' interpretation of items [Brislin, 1970]. In the context of game studies, user experience is a multidimensional construct encompassing factors such as immersion, engagement, competence, and affect, influenced by cultural and individual variables. The Game Experience Questionnaire (GEQ), developed by Poels, de Kort, and IJsselstein [2007], is one of the most widely used instruments to evaluate these dimensions, applied in diverse contexts from entertainment games to serious health [IJsselstein *et al.*, 2008] and education interventions.

The GEQ is organized into modules, each focused on different aspects of the player's experience:

- **Core Module (CM):** Comprising 33 items, theoretically conceived to measure player experience across 7 constructs: competence, immersion, flow, tension, challenge, positive affect, and negative affect. Empirical analyses frequently reveal instabilities in its factor structure, suggesting better fit with a 5-factor model by aggregating tension, challenge, and negative affect, and excluding items Q23, Q32, and Q5. Such instabilities have also been reported in international studies.
- **Post-Game Module (PGQ):** Designed to measure the dimensions of Positive and Negative Experience, Tiredness, and Return to Reality. Compatibility of factors with theoretical constructs is partial, but there is substantial correspondence (more than 80%) of items, with the analysis indicating a 4-out-of-5 factor structure.
- **Social Presence Module (SPGQ):** Covers the dimensions of Empathy, Negative Feeling, and Involvement. Similarly to the PGQ, compatibility with theoretical constructs is partial, but with correspondence of 2 out of 3 factors and more than 80% of items.

For a correct interpretation of GEQ results and those

of similar instruments, it is essential to distinguish between *constructs*, *factors*, and *components*. Constructs refer to the theoretical, abstract concepts the GEQ proposes to measure, representing the underlying dimensions of player experience conceptualized in its theoretical foundation (e.g., immersion, competence, positive affect). Factors are the latent dimensions or item clusters that emerge empirically from statistical analyses such as EFA, indicating the observable manifestation of those constructs in collected data. Finally, in the context of Principal Component Analysis (a common EFA method), components are linear combinations of observed variables (questionnaire items) extracted to explain maximum variance in the data, often used analogously to factors. This distinction is crucial because the empirically derived factor or component structure does not always perfectly correspond to the theoretical constructs originally proposed.

The scientific literature presents a variety of questionnaires designed to measure different aspects of game experience, varying in scope from general player satisfaction assessments to more specific measures of immersion, flow, competence, emotions, and social presence.

The Player Experience of Need Satisfaction (PENS) is a model grounded in Self-Determination Theory that evaluates the satisfaction of basic psychological needs — autonomy, competence, and relatedness — in the context of digital games, providing a theoretical framework for understanding the motivational factors underlying player experience [Johnson *et al.*, 2018]. Complementarily, the Player Experience Inventory (PXI) was developed as a robust, theoretically grounded tool that allows researchers to systematically analyze how game design choices influence player experience, offering reliable metrics for empirical studies [Abeele *et al.*, 2020].

In the gamification domain, the Gameful Experience Questionnaire (GAMEFULQUEST) emerges as a validated instrument to measure gamified experience in non-game contexts, evaluating how game elements are employed to increase engagement and motivation in various services and applications [Högberg, 2019]. Notably, Oliveira da Silva Junior *et al.* [2024] adapted and validated GAMEFULQUEST for Brazilian Portuguese. Although not game-specific, the Intrinsic Motivation Inventory (IMI) is frequently adapted to measure intrinsic motivation in ludic activities. The System Usability Scale (SUS) [Lourenço *et al.*, 2022] is widely used as a generic system evaluation measure, applicable to games when appropriately adapted. Finally, the Video Game Experience Measure (VGEM) [Williams *et al.*, 2023] stands out as a generalized measure of video game experience, capable of discriminating non-players from players and sensitive to different proficiency levels, making it useful for comparative studies.

The specialized literature highlights that cultural adaptation of questionnaires such as the GEQ requires a systematic and rigorous approach ensuring semantic, idiomatic, conceptual, and cultural equivalence between the original and adapted versions. Beaton *et al.* [2000] propose methodological guidelines for this process, including initial translation, back-translation, expert committee review, and pilot testing. In the case of the GEQ, adaptation into Brazilian Portuguese required special attention to idiomatic expressions,

regional variations, and cultural differences that may affect Brazilian players' interpretation of items. Previous studies have demonstrated the GEQ's applicability across different cultural contexts, highlighting its utility for evaluating game experience in health interventions, such as exergames and serious games. Hoffmann et al. [2015] used the GEQ to investigate the effectiveness of personalized exercise games, while Boletis and McCallum [2016] applied the instrument in a serious game for cognitive screening of elderly individuals. In Brazil, the absence of a validated GEQ version has limited comparative studies and local applications, particularly in health, where user experience is critical to intervention success.

3 Participants, Methods, and Procedures

3.1 Participants

Research participants were recruited from students at a private university in São Paulo, Brazil. The sample consisted of 115 players.

All participants were informed of the research objectives and signed a free and informed consent form (TCLE) prior to participation. Confidentiality and anonymity were guaranteed in accordance with ethical research guidelines. The study was approved by the Institutional Human Research Ethics Committee.

3.1.1 Inclusion criteria

- Experience with electronic games:** participants with regular experience playing electronic games, demonstrating familiarity with the gaming environment and its mechanics.
- Availability to participate:** availability to take part in all stages of the research, including the testing phase and completion of the questionnaire.
- Age:** being over 18 years of age, ensuring the legal capacity to consent to research participation.
- University students:** being a student of various programs at the host university, enabling the integration of the research within the academic environment.
- Interest in education- and health-related games:** interest in serious games developed at the university, focused on education and health, ensuring the relevance of the sample to the study's objective.
- Agreement with the research terms:** signing the free and informed consent form (TCLE), agreeing to participate in the study and to share the necessary information.

3.1.2 Exclusion criteria

- Lack of experience with electronic games:** participants without regular or minimal experience with electronic games, which could compromise the relevance of their responses to the research.
- Under 18 years of age:** participants younger than 18, due to the need for legal consent and the nature of the study.
- Conflict of interest:** participants directly involved with the development or translation of the Game Experience Questionnaire (GEQ), which could introduce bias into

the responses.

- Inability to play:** any condition preventing the participant from interacting adequately with the selected electronic games, such as severe motor difficulties or uncorrected vision problems.
- Participants not enrolled at the university:** individuals who are not students at the host university, since the study is restricted to the university environment.
- Non-agreement with the research terms:** refusal to sign the free and informed consent form (TCLE), preventing participation in the study.

Table 1 presents the characterization of the 115 respondents across the three GEQ modules (CM = Core Module, PGQ = Post-Game, SPGQ = Social Presence). Most respondents are university students (63%), aged 18–24 (83%), with regular gaming use (36%) and a preference for PC games (56%).

Table 1. Sample characterization of respondents (n = 115).

	Female	Male	N/A	All
<i>Age</i>				
18–24 years	0.72	0.86	1.00	0.83
25–34 years	0.24	0.11	0.00	0.14
35–44 years	0.04	0.02	0.00	0.03
<i>Education level</i>				
Primary complete	0.00	0.02	0.00	0.02
Secondary complete	0.16	0.29	0.33	0.26
Higher complete	0.00	0.10	0.00	0.08
Higher incomplete	0.84	0.57	0.67	0.63
Post-graduation	0.00	0.01	0.00	0.01
<i>Gaming frequency</i>				
Every day	0.12	0.28	0.00	0.23
Never	0.12	0.01	0.00	0.03
Occasionally	0.28	0.20	0.67	0.23
Rarely	0.36	0.09	0.00	0.15
Regularly	0.12	0.43	0.33	0.36
<i>Preferred device</i>				
Other	0.04	0.01	0.00	0.02
PC	0.24	0.66	0.33	0.56
PlayStation	0.08	0.15	0.00	0.13
Smartphone	0.52	0.15	0.67	0.24
Switch	0.00	0.01	0.00	0.01
Tablet	0.08	0.00	0.00	0.02
Xbox	0.04	0.02	0.00	0.03

Research participants were players with regular experience playing electronic games. The sample was recruited individually and voluntarily, ensuring a diverse representation of players with varied interests across game genres. The games selected for this study focus on serious games developed at the university itself, related to the fields of education and health, by convenience.

3.2 Stages

The cultural adaptation process of the GEQ into Brazilian Portuguese followed a methodological approach based on the guidelines proposed by Beaton et al. [2000] for cross-cultural adaptation of research instruments, divided into the following sequential stages.

3.2.1 Stage 1 – Initial Translation

The first stage consisted of translating the original English GEQ into Brazilian Portuguese. This translation was performed by an independent translator, native in Portuguese with advanced English proficiency, who possessed conceptual knowledge of the instrument and prior experience in academic material translation. The initial translation sought to preserve the literal meaning of items while considering possible cultural and linguistic nuances of the Brazilian target audience.

3.2.2 Stage 2 – Back-Translation

After the initial translation, the questionnaire was back-translated into English by a second translator, also fluent in both languages but without access to the original GEQ. This back-translation step is essential for identifying discrepancies between original and translation, ensuring the meaning of items has been preserved. The comparison between back-translation and original allowed identification of possible semantic or conceptual deviations, which were subsequently adjusted.

3.2.3 Stage 3 – Synthesis of Translations

The two translations (initial and back-translation) were compared and synthesized into a single Portuguese version by an impartial third party with experience in research methodology, resolving inconsistencies and ensuring the original meaning of items was maintained.

3.2.4 Stage 4 – Expert Committee Review

The preliminary version was submitted for evaluation by a multidisciplinary committee composed of two methodologists experienced in validation research, two linguists, two professionals from the field of Human-Computer Interaction and digital game development, and representatives from the target audience (players), in addition to the translators. The committee evaluated semantic, idiomatic, conceptual, and cultural equivalence of items, ensuring the adapted version was comprehensible and relevant to the Brazilian context.

3.2.5 Stage 5 – Pilot Test

The pre-final version of the adapted GEQ was tested by the participants. After the game session, participants responded to the adapted GEQ using a five-point Likert scale (0 = “Not at all” to 4 = “Extremely”), following the original model.

3.2.6 Stage 6 – Data Analysis

Data collected in the pilot test were analyzed using statistical techniques appropriate for research instrument validation. Internal consistency was assessed through Cronbach's alpha (α), with a desired minimum value of 0.8. Additionally, Exploratory Factor Analysis (EFA) was performed using the principal components method with oblique Promax rotation in all cases, given that GEQ constructs are expected to be correlated. The Kaiser-Meyer-Olkin (KMO) index and Bartlett's sphericity test verified data adequacy for EFA.

3.2.7 Stage 7 – Final Adjustments

Based on pilot test and data analysis results, final adjustments were made to the adapted GEQ, including reformulation of items with low factor loadings or comprehension difficulties, and revision of terms that could generate ambiguities.

There are several indices for measuring the reliability or internal consistency of a scale. Cronbach's [1951] is a traditionally used index, but McDonald's Omega (ω) is a more recent estimator that has been widely applied [Hayes and Coutts, 2020].

4 Results

Cronbach's alpha indicated high internal consistency, reinforcing the reliability of the adapted scale for evaluating game experience in the Brazilian context.

Table 2 shows the results of applying EFA across the three GEQ modules for the 115 respondents, a sample considered acceptable according to Fabrigar and Wegener [2012]. For principal component analysis, the oblique Promax rotation method was used in all cases. The KMO index and Bartlett's test of sphericity were adequate: Bartlett's test was significant for all three modules ($p < 0.05$) and KMO showed sample adequacy [Kaiser, 1974] ranging from good (0.7–0.8) to meritorious (0.8–0.9).

4.1 GEQ Core Module

Table 3 presents the rotated standard pattern matrix (Promax) for the seven factors: competence, immersion, flow, tension, challenge, positive affect, and negative affect. Compared to the original theoretical GEQ constructs, the Challenge and Negative Affect components show no correspondence in the empirical data, presenting low loadings (< 0.4), cross-loadings, or divergence from the main component, suggesting better adequacy with a 5-factor model. A similar lack of correspondence between the theoretical model and its empirical application was found by Johnson et al. [2018] and Law et al. [2018], reaching similar results suggesting a 5-factor structure (aggregating tension, challenge, and negative affect) and excluding items Q23, Q32, and Q5, which also find correspondence in our analysis. Regarding communalities, only Q8 and Q16 present values < 0.4 .

4.2 PGQ Post-Game and SPGQ Social Presence

The EFA summary for the Post-Game (PGQ) and Social Presence (SPGQ) modules is presented in Tables 4 and 5. Similarly to the CM, the compatibility of factors with theoretical constructs is partial, but there is substantial correspondence of more than 80% of items in both cases, with component correspondence of 4 out of 5 factors for the PGQ and 2 out of 3 for the SPGQ, with mean communalities of 0.55 and 0.59 respectively.

4.3 Internal Reliability

To assess the internal consistency of the adapted GEQ modules, reliability analyses were performed using both Cronbach's alpha (α) and McDonald's omega (ω), as detailed in Section 3.2.7. Table 6 presents the reliability coefficients obtained for each module and their respective subscales.

For both indices, values > 0.7 are considered acceptable (consistent measures), with values > 0.8 being ideal. All modules present reliability indices from acceptable to ideal on both Cronbach's and ω metrics. For components, only Negative Affect (GEQ), Return to Reality (PGQ), and Negative Feeling (SPGQ) present values below the accept-

Table 2. EFA sample adequacy: KMO and Bartlett's test.

Module	KMO	Adequacy	Bartlett χ^2	<i>p</i>
GEQ <i>Core</i>	0.844	Meritorious (0.8–0.9) ¹	2427.12	2.1×10^{-240}
PGQ <i>Post-G</i>	0.771	Good (0.7–0.8)	921.00	8.8×10^{-117}
SPGQ <i>Soc.</i>	0.842	Meritorious (0.8–0.9)	1331.21	3.7×10^{-195}

¹ Scale [Kaiser, 1974]: >0.90 Excellent; 0.8–0.9 Meritorious; 0.7–0.8 Good; 0.6–0.7 Mediocre; <0.5 Unacceptable.

Table 3. GEQ Core Module — Rotated pattern matrix (Promax) for 33 items on 7 theoretical factors.

Item	Brazilian Portuguese (abbreviated)	F1	F2	F3	F4	F5	F6	F7	Note	h2
<i>Competence (F3)</i>										
Q2	Eu me senti habilidoso	0.03	-0.04	0.78	0.08	0.05	0.12	-0.05		0.64
Q10	Eu me senti competente	0.04	-0.27	0.64	0.06	-0.07	0.10	0.28		0.64
Q15	Fui bom nele	0.15	0.13	0.82	0.06	-0.06	0.00	-0.24		0.75
Q17	Eu me senti bem-sucedido	0.06	-0.09	0.70	0.15	-0.04	0.05	0.18		0.64
Q21	Fui rápido em atingir os obj.	0.24	0.15	0.70	-0.04	-0.02	0.01	-0.27		0.63
<i>Immersion (F5)</i>										
Q3	Fiquei interessado na história	0.06	0.09	-0.05	0.08	0.79	-0.15	-0.14		0.67
Q12	Era esteticamente agradável	0.39	-0.05	0.02	-0.20	0.44	0.07	0.17		0.45
Q18	Eu me senti imaginativo	0.03	-0.02	0.20	0.29	0.54	0.05	-0.07		0.64
Q19	Senti que podia explorar as coisas	0.00	-0.12	0.16	0.19	0.76	-0.06	0.11		0.66
Q27	Achei impressionante	0.27	0.05	-0.05	0.11	0.51	-0.09	0.24		0.66
Q30	Pareceu uma experiência valiosa	0.14	-0.10	0.14	0.49	0.38	-0.03	-0.24	<i>diverges</i>	0.62
<i>Flow (F4)</i>										
Q5	Eu estava totalmente envolvido	0.61	0.05	0.16	0.12	0.05	0.12	-0.10	<i>diverges</i>	0.59
Q13	Esqueci de tudo ao meu redor	-0.03	0.09	0.00	0.75	-0.01	-0.04	-0.14		0.66
Q25	Perdi a noção do tempo	-0.08	0.18	0.11	0.61	0.33	-0.11	-0.09		0.63
Q28	Estava profundamente concentrado	0.29	0.04	0.13	0.59	-0.05	-0.03	0.04		0.69
Q31	Perdi a conexão c/ mundo exterior	-0.35	-0.06	0.17	0.94	0.17	0.19	-0.12		0.78
<i>Tension (F2)</i>										
Q22	Eu me senti incomodado	-0.06	0.77	0.13	0.04	-0.08	0.07	-0.07		0.63
Q24	Eu me senti irritado	0.15	0.81	-0.12	0.01	0.12	-0.11	0.03		0.65
Q29	Eu me senti frustrado	0.18	0.70	-0.26	0.05	-0.15	0.18	0.03		0.75
<i>Challenge — instability noted</i>										
Q11	Achei ele difícil	0.06	0.17	-0.30	0.09	0.06	0.11	0.57	<i>diverges</i>	0.60
Q23	Eu me senti pressionado	0.12	0.43	-0.07	0.06	-0.21	0.36	0.06	<i>diverges</i>	0.46
Q26	Eu me senti desafiado	0.09	0.00	-0.03	0.58	-0.10	0.22	0.17	<i>div.</i>	0.57
Q32	Senti a pressão do tempo	0.01	-0.12	0.17	0.12	-0.09	0.88	0.03	<i>diverges</i>	0.70
Q33	Tive de me esforçar muito	-0.04	0.09	-0.15	0.18	0.27	0.32	0.31	<i>weak</i>	0.56
<i>Negative Affect — instability noted</i>										
Q7	Isso me deixou de mau humor	0.04	0.77	-0.06	0.11	0.15	-0.08	0.05		0.67
Q8	Pensei em outras coisas	-0.10	0.20	0.20	-0.25	0.39	0.20	0.08	<i>weak</i>	0.33
Q9	Achei ele cansativo	-0.15	0.56	0.04	-0.14	-0.06	0.01	-0.19	<i>cross</i>	0.57
Q16	Eu me senti entediado	-0.28	0.49	0.16	-0.08	-0.01	-0.13	0.16		0.39
<i>Positive Affect (F1)</i>										
Q1	Eu me senti contente	0.58	0.05	0.28	0.12	-0.11	-0.20	0.07		0.59
Q4	Achei divertido	0.82	-0.01	0.05	-0.28	0.21	0.07	-0.10		0.65
Q6	Eu me senti feliz	0.76	0.07	0.11	-0.06	-0.03	0.05	0.11		0.60
Q14	Eu me senti bem	0.61	-0.05	0.13	0.15	-0.03	-0.14	0.29		0.72
Q20	Gostei dele	0.58	-0.09	-0.01	-0.03	0.40	-0.06	-0.16		0.71
Sum of Variances		5.21	4.90	4.46	5.25	4.53	1.73	2.25	–	mean
% Variances		18.40	17.30	15.72	18.52	16.00	6.11	7.95	–	0.62

Promax rotation. **Bold:** primary loading (≥ 0.4 on expected factor). *weak:* loading < 0.4 ; *cross:* cross-loading; *diverges:* diverges from theoretical construct. *h2* = communality; mean at column end.

able threshold — precisely the same components that showed the greatest divergence in the EFA. It is also worth noting that the CM components with the lowest reliability values (Negative Affect and Challenge) are the same ones presenting unacceptable values in the review by Law et al. [2018].

In summary, respondent data constituted an adequate sample for EFA, with KMO > 0.7 and significant Bartlett's test for all three GEQ modules. The EFA showed acceptable adherence of the translated Portuguese questionnaire structure to the original GEQ, presenting — as expected — sim-

Table 4. PGQ Post-Game Module — Rotated pattern matrix (Promax) for 17 items on 4 theoretical factors.

Item	Construct	F1	F2	F3	F4	h2
<i>Positive Experience (F1)</i>						
Q1	Pos. Exp.	0.75	-0.06	0.06	0.24	0.68
Q5	Pos. Exp.	0.73	0.11	-0.13	-0.04	0.57
Q7	Pos. Exp.	0.74	-0.16	0.28	0.13	0.60
Q8	Pos. Exp.	0.68	-0.17	-0.07	0.17	0.62
Q12	Pos. Exp.	0.78	0.09	-0.01	-0.28	0.63
Q16	Pos. Exp.	0.84	-0.06	0.08	-0.06	0.67
<i>Negative Experience (F2)</i>						
Q2	Neg. Exp.	-0.13	0.58	0.18	0.12	0.47
Q4	Neg. Exp.	0.05	0.80	-0.03	0.70	<i>cross</i>
Q6	Neg. Exp.	-0.19	0.53	0.19	0.04	0.46
Q11	Neg. Exp.	-0.09	0.41	0.05	0.10	0.19
Q14	Neg. Exp.	0.16	0.84	-0.20	0.01	0.60
Q15	Neg. Exp.	-0.01	0.46	0.19	-0.25	0.48
<i>Tiredness (F3)</i>						
Q10	Tiredness	0.15	-0.11	0.97	0.06	0.82
Q13	Tiredness	0.13	0.05	0.70	-0.18	0.57
<i>Return to Reality — divergence noted</i>						
Q3	Ret. Real.	0.27	0.60	-0.13	0.19	<i>diverges</i>
Q9	Ret. Real.	-0.04	0.32	0.45	0.06	<i>diverges</i>
Q17	Ret. Real.	0.63	0.19	0.06	-0.06	<i>diverges</i>
Sum of Variances		5.45	4.40	2.65	0.93	mean
% Variances		40.57	32.77	19.75	6.90	0.55

Promax rotation. **Bold**: primary loading. weak: loading <0.4; cross: cross-loading; diverges: diverges from theoretical construct. h2 = communality.

Table 5. SPGQ Social Presence Module — Rotated pattern matrix (Promax) for 17 items on 3 theoretical factors.

Item	Construct	F1	F2	F3	h2
<i>Empathy (F1)</i>					
Q1	Empathy	0.46	0.12	-0.26	0.31
Q4	Empathy	0.40	0.39	-0.00	0.53
Q8	Empathy	0.70	-0.07	0.12	0.47
Q9	Empathy	0.98	-0.12	-0.10	0.79
Q10	Empathy	0.89	0.01	-0.14	0.79
Q13	Empathy	0.64	0.11	0.11	0.55
<i>Negative Feeling (F3) — divergence noted</i>					
Q7	Neg. Feel.	-0.10	0.21	0.66	0.52
Q11	Neg. Feel.	0.76	-0.08	-0.01	<i>diverges</i>
Q12	Neg. Feel.	0.73	-0.07	0.22	<i>diverges</i>
Q16	Neg. Feel.	0.02	0.02	0.70	0.50
Q17	Neg. Feel.	0.03	-0.21	0.92	0.78
<i>Involvement (F2)</i>					
Q2	Involvement	-0.16	0.96	-0.13	0.68
Q3	Involvement	-0.18	0.91	-0.01	0.63
Q5	Involvement	0.36	0.51	-0.07	0.62
Q6	Involvement	0.17	0.59	0.12	0.57
Q14	Involvement	0.23	0.58	0.10	0.61
Q15	Involvement	0.15	0.63	0.15	0.63
Sum of Variances		6.08	4.49	2.36	mean
% Variances.		46.99	34.74	18.27	0.59

Promax rotation. **Bold**: primary loading. weak: loading <0.4; cross: cross-loading; diverges: diverges from theoretical construct. h2 = communality.

ilar factor structure instabilities to the original, already reported in previous studies [Johnson *et al.*, 2018; Law *et al.*, 2018]. The translation generally shows good reliability with Cronbach's and $\omega > 0.7$. In other words, the cross-cultural Portuguese translation is an adequate scale for the measures it proposes, sharing both the qualities and limitations of the

original scale.

Our analysis confirms that the GEQ scale, as demonstrated by Johnson *et al.* [2018] and Law *et al.* [2018] for the core module, presents certain factor structure instabilities, and contributes here by demonstrating similar instabilities in the Post-Game and Social Presence modules as well.

Table 6. Internal reliability: ω , Cronbach's α , and 95% CI.

Module / Component	ω	α	95% CI
GEQ*	0.993	0.911	[0.886, 0.933]
Competence	0.963	0.870	[0.828, 0.904]
Immersion	0.955	0.875	[0.836, 0.907]
Flow	0.963	0.857	[0.811, 0.894]
Tension	0.948	0.795	[0.720, 0.852]
Challenge	0.945	0.768	[0.694, 0.829]
Negative Affect	0.936	0.672	[0.562, 0.760]
Positive Affect	0.958	0.870	[0.828, 0.904]
PGQ*	0.978	0.812	[0.758, 0.859]
Positive Experience	0.953	0.884	[0.848, 0.914]
Negative Experience	0.932	0.775	[0.705, 0.833]
Tiredness	0.919	0.790	[0.696, 0.855]
Return to Reality	0.833	0.394	[0.173, 0.563]
SPGQ*	0.973	0.910	[0.884, 0.932]
Empathy	0.921	0.872	[0.833, 0.905]
Negative Feeling	0.918	0.694	[0.596, 0.775]
Involvement	0.933	0.896	[0.864, 0.923]

* All components combined. **Bold:** $\alpha < 0.7$.

Nonetheless, the scale remains an acceptable measurement instrument for several important dimensions of digital game experience and should continue as the predominant measurement instrument — which underscores the importance of a cross-cultural translation into Brazilian Portuguese.

Despite factorial instabilities observed across all modules, the GEQ remains an acceptable measurement instrument for several important dimensions of digital game experience and is considered the predominant tool for such assessment. The Brazilian Portuguese translation demonstrated general reliability, with indices ranging from acceptable to ideal, although certain specific components (Negative Affect in the CM, Return to Reality in the PGQ, and Negative Feeling in the SPGQ) presented values below the acceptability threshold.

5 Brazilian Portuguese Version of the GEQ

The following tables present the final Brazilian Portuguese version of the three GEQ modules after all stages of translation and cultural adaptation. The final version of the GEQ Core Module is defined in Table 7.

The final version of the GEQ Post-Game Module is defined in Table 8.

The final version of the GEQ Social Presence Module is defined in Table 9.

6 Discussion

The detailed EFA results for the GEQ Core Module corroborate observations from previous studies [Johnson *et al.*, 2018; Law *et al.*, 2018], which pointed to instabilities in the theoretical factor structure of the instrument. Specifically, the present analysis revealed that the “Challenge” and “Negative Affect” components showed no robust empirical correspondence, presenting low factor loadings (< 0.4), significant cross-loadings, or divergence from the main component. This finding suggests that, although the original theoretical GEQ model postulates seven factors, empirical ap-

Table 7. Brazilian version of the GEQ Core Module.

#	Brazilian Portuguese
1	Eu me senti contente
2	Eu me senti habilidoso
3	Fiquei interessado na história do jogo
4	Achei divertido
5	Eu estava totalmente envolvido com o jogo
6	Eu me senti feliz
7	Isso me deixou de mau humor
8	Pensei em outras coisas
9	Achei ele cansativo
10	Eu me senti competente
11	Achei ele difícil
12	Era esteticamente agradável
13	Esqueci de tudo ao meu redor
14	Eu me senti bem
15	Fui bom nele
16	Eu me senti entediado
17	Eu me senti bem-sucedido
18	Eu me senti imaginativo
19	Senti que podia explorar as coisas
20	Gostei dele
21	Fui rápido em atingir os objetivos do jogo
22	Eu me senti incomodado
23	Eu me senti pressionado
24	Eu me senti irritado
25	Perdi a noção do tempo
26	Eu me senti desafiado
27	Achei impressionante
28	Estava profundamente concentrado no jogo
29	Eu me senti frustrado
30	Pareceu uma experiência valiosa
31	Perdi a conexão com o mundo exterior
32	Senti a pressão do tempo
33	Tive de me esforçar muito no jogo

Table 8. Brazilian version of the Post-Game GEQ.

#	Brazilian Portuguese
1	Eu me senti revigorado
2	Eu me senti mal
3	Achei difícil voltar à realidade
4	Eu me senti culpado
5	Foi como uma vitória
6	Achei uma perda de tempo
7	Eu me senti energizado
8	Eu me senti satisfeito
9	Eu me senti desorientado
10	Eu me senti exausto
11	Senti que poderia ter feito coisas mais úteis
12	Eu me senti poderoso
13	Eu me senti esgotado
14	Senti remorso
15	Eu me senti envergonhado
16	Eu me senti orgulhoso
17	Tive a sensação de ter voltado de uma viagem

plication with Brazilian players tends toward better fit with a five-factor structure, similarly to what was observed in other populations. The consistency of these instabilities across dif-

Table 9. Brazilian version of the Social Presence GEQ.

#	Brazilian Portuguese
1	Tive empatia com o(s) outro(s)
2	As minhas ações dependiam das ações do(s) outro(s)
3	As ações dos outros dependiam das minhas ações
4	Eu me senti conectado com o(s) outro(s)
5	Os outros prestaram atenção em mim
6	Prestei muita atenção no(s) outro(s)
7	Senti ciúmes do(s) outro(s)
8	Achei que foi agradável estar com o(s) outro(s)
9	Quando eu estava feliz, o(s) outro(s) também estava(m) feliz(es)
10	Quando o(s) outro(s) estava(m) feliz(es), eu também estava feliz
11	Influenciei o humor do(s) outro(s)
12	Fui influenciado pelo humor do(s) outro(s)
13	Admirei o(s) outro(s)
14	O que o(s) outro(s) fazia(m) afetava o que eu fazia
15	O que eu fazia afetava o que o(s) outro(s) fazia(m)
16	Eu me senti vingativo
17	Senti tristeza

ferent cultural and linguistic contexts reinforces the need for a deeper review of the GEQ's psychometric structure to ensure its cross-cultural validity [Johnson *et al.*, 2018; Law *et al.*, 2018].

Additionally, investigation of the factor structure of the PGQ and SPGQ modules revealed analogous instability patterns to those identified in the CM. Although compatibility of factors with original theoretical constructs is partial, substantial correspondence of more than 80% of items in both modules was observed — 4 out of 5 factors for the PGQ and 2 out of 3 for the SPGQ, with mean communalities of 0.55 and 0.59 respectively. These results are crucial because they extend the understanding of the GEQ's structural limitations beyond the core module, evidencing that factorial instability is a characteristic that permeates the different components of the questionnaire, regardless of application context [Johnson *et al.*, 2018; Law *et al.*, 2018].

The internal reliability analysis, employing both Cronbach's Alpha and McDonald's Omega (ω) [Cronbach, 1951; Hayes and Coutts, 2020], demonstrated that the adapted GEQ modules exhibit indices from acceptable to ideal, with values consistently above 0.7. However, three specific components — Negative Affect (GEQ), Return to Reality (PGQ), and Negative Feeling (SPGQ) — presented values below the acceptability threshold, correlating with the structural instability identified in the EFA. These findings underscore the complexity inherent in measuring game experience and the importance of rigorous psychometric validation across different cultural contexts.

The persistence of factorial instabilities in the GEQ structure, even after a meticulous cross-cultural adaptation process, suggests that some constructs may be more sensitive to cultural variation, or that the theoretical conceptualization of certain game experience dimensions may itself require refinement. The replication of these instabilities in Brazilian studies, consistent with international findings, reinforces the ecological validity of the results and the robustness of the

methodology employed [Law *et al.*, 2018].

Given the observed instabilities, it is essential that researchers and game developers in Brazil consider the practical implications of these findings. Although the GEQ remains a valuable instrument for game experience evaluation, interpretation of its results should be conducted with caution, especially regarding dimensions with lesser factorial stability and internal consistency. Complementary approaches — such as qualitative methods or combination with other game experience scales — are recommended for a more comprehensive and nuanced understanding of the Brazilian player experience [Johnson *et al.*, 2018].

In summary, the Brazilian Portuguese GEQ, while presenting factor structure instabilities similar to those reported in international studies, remains an acceptable measurement instrument for several crucial dimensions of digital game experience. Its general reliability, with Cronbach's and $\omega > 0.7$, attests its adequacy for the proposed measures, sharing both the qualities and limitations of the original scale. This study not only validates the cross-cultural translation of the GEQ into Brazilian Portuguese, but also contributes to the literature by demonstrating the persistence of certain structural instabilities in less-explored modules — Post-Game and Social Presence — reinforcing the importance of an in-depth understanding of psychometric properties of research instruments across different cultural contexts [Law *et al.*, 2018; Johnson *et al.*, 2018].

Despite its widespread application, the GEQ has faced challenges regarding its formal validation and consistency of factor structure. A systematic review of 73 publications using the GEQ revealed inconsistencies in how the questionnaire is cited and reported, and a lack of evidence for the original 7-factor structure [Law *et al.*, 2018]. Validation studies, such as that of Johnson *et al.* [2018], indicate that while the GEQ is partially supported, a revised factor structure may be necessary. Brühlmann and Schmid [2015] also explored the GEQ's factor structure, finding support for the PENS structure but unable to verify the proposed GEQ structure. These findings underscore the importance of careful and informed application of the GEQ, especially given its empirically unstable factor structure.

6.1 Limitations and Future Work

Despite the promising psychometric results, some limitations of this study must be acknowledged. First, the sample size of 115 participants is relatively modest for a multidimensional validation of an instrument with three modules totaling 67 items. Although guidelines for Exploratory Factor Analysis (EFA) indicate that smaller samples can yield reliable factor structures when communalities are high and factor loadings are strong [Fabrigar and Wegener, 2012], the current sample size remains a limitation. Second, because this research relied on a convenience sample of university students playing serious games, caution should be exercised when generalizing these findings to the broader, more diverse Brazilian gaming population. Future studies should employ larger and more demographically diverse samples across different regions of Brazil, enabling the use of Confirmatory Factor Analysis (CFA) to further consolidate and test the structural validity of this adapted version of the GEQ.

7 Conclusion

Study results indicate that the adapted GEQ version for Brazilian Portuguese is valid and reliable for evaluating the game experience of Brazilian players. The EFA confirmed the overall modular structure of the original instrument (Core, Post-Game, and Social Presence); however, in line with prior international validation studies, it also revealed specific factor structure instabilities, particularly in the challenge and negative affect components. Regarding internal consistency, reliability analyses demonstrated high reliability for all three overall modules, with McDonald's omega (ω) coefficients exceeding 0.90 and Cronbach's alpha (α) values above 0.80. Nonetheless, certain specific subscales (such as Negative Affect in the Core Module, Return to Reality in the Post-Game Module, and Negative Feeling in the Social Presence Module) exhibited lower reliability, closely mirroring the empirical challenges reported in the literature. The results of this research demonstrated that the adapted GEQ version for Brazilian Portuguese showed semantic, cultural, and conceptual equivalence with the original instrument.

The cultural adaptation of the GEQ faced challenges related to Brazil's linguistic and cultural diversity. The selection of terms capturing cultural nuances without sacrificing fidelity to original concepts was a constant concern during the translation process. Expert review and pilot testing were essential for identifying and correcting possible ambiguities and misunderstandings in questionnaire items.

The availability of the GEQ adapted for Brazilian Portuguese has significant practical and theoretical implications. For academic researchers, the validated instrument enables comparative studies across different cultures and the development of more inclusive theories about game experience. For the games industry, the adapted version facilitates systematic evaluation and continuous improvement of user experience in games developed for the Brazilian market.

Future studies should include more diversified samples covering different age groups, educational levels, and geographic regions of Brazil, in order to verify the stability of the instrument's psychometric properties in broader contexts. Additionally, advanced statistical techniques such as Confirmatory Factor Analysis (CFA) and Structural Equation Modeling (SEM) are recommended to more rigorously test the proposed factor structure and investigate causal relationships among game experience constructs. Longitudinal studies could also examine the temporal stability of GEQ measures, contributing to a more dynamic and comprehensive understanding of the construct.

This study contributes to advancing game studies in Brazil by providing a validated and culturally sensitive version of the GEQ for Brazilian Portuguese, opening new possibilities for comparative research across different cultures and supporting the development of games better aligned with the preferences and needs of the Brazilian audience.

Declarations

Authors' Contributions

The authors' contributions, following the CRediT taxonomy, are described below.

- Douglas Fabiano Lourenço: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Project administration, Writing – original draft.
- Rogério de Oliveira: Conceptualization, Methodology, Supervision, Validation, Writing – review & editing.
- Silvana Maria Blascovi de Assis: Formal analysis, Investigation, Supervision, Validation, Writing – review & editing.
- Ana Grasielle Dionísio Corrêa: Conceptualization, Supervision, Validation, Writing – review & editing.

Competing interests

The authors declare no competing interests.

Availability of data and materials

The Brazilian Portuguese versions of the GEQ Core Module, Post-Game Questionnaire, and Social Presence Questionnaire, together with the anonymized pilot-study data and the statistical analysis code, are openly available in the project repository on GitHub: <https://github.com/GEQ-Portuguese-Adaptation/geq-portuguese-adaptation>. An archived version of the repository and associated research materials is available through Zenodo at <https://doi.org/10.5281/zenodo.22681094>.

Further relevant information

This research was approved by the Institutional Human Research Ethics Committee of the host university (CAAE:83094424.6.0000.0084). All procedures followed the ethical standards of the Brazilian National Health Council (CNS Resolution 510/2016). All participants provided written informed consent prior to participation.

The authors declare that AI-based writing assistance tools (ChatGPT, OpenAI) were used for language editing, grammatical revision, and structural improvements of the English text in this manuscript. All scientific content, research design, data collection, analysis, interpretation, and conclusions are exclusively the authors' intellectual work. Any use of AI has been fully disclosed as required by the journal.

Deliberate efforts were made to incorporate the work of Brazilian and Latin American researchers, taking into account the importance of diversity and inclusion in citations.

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